

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000832		
	Version:	5.0; Most-Recent; Effective; CURRENT		
	Status:	Effective	Purpose:	Uncontrolled Copy
Class:	Control del Diseño		Effective date:	24-Apr-2018
Title:	SM DG57-DG517. Centrifuge verification procedure.			

DG-57 / DG-517 Service Manual

CENTRIFUGE VERIFICATION PROCEDURE

Procedure description
This document describes the procedure that should be followed in order to verify whether the centrifuge module meets the necessary conditions to guarantee a proper centrifugation of gel cards. Note: <i>Document REGD-0000832_v5 replaces document REGD-0000832_v4.</i>

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EQUIPMENT

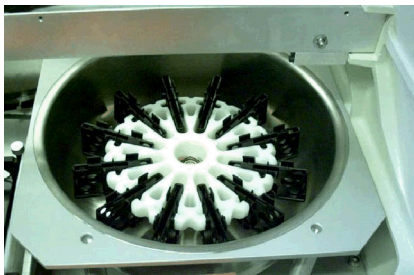
1. A digital photo tachometer with accuracy of 0.1% @ 1000 RPM (e.g. Lutron DT-2236).
2. Reflective tape (8 x 20 mm)
3. A chronometer having at least a precision about 1 second in 10 minutes.

SET UP

1. Turn off the instrument
2. See Document REGD-0000804

	NOTE: Some phototachometers (e.g. referenced one in equipment section) have mechanical coupling option. Do not use that option. Do not put anything through the centrifuge aperture during the test.
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PROCEDURE

Process	Expected Results
1. Open the Upper Door. 2. Put a piece of Reflective Tape on the Centrifuge Rotor, between two card holders. The tape should not prevent the holders free movement. This can be done through the centrifuge aperture, provided it is not moving.  Picture 1	---
3. Execute the ST software (Technical Service software). 4. Enter the Technical Service password (d+g_diana). 5. Click on <i>Centrifuge/centrifugation</i> and set the <i>R.P.M</i> parameter at 1228. Then set the <i>Seconds</i> parameter at 540. Caution: The centrifuge will be moving during the following steps. Avoid introducing any object in the module. 6. Click on <i>OK</i> and start the time measure with the chronometer at the same time. 7. The centrifuge will reach nominal speed in a few seconds. Then focus the phototachometer light on the reflecting tape. Read the display of the tachometer and compare the returned value with the expected result. 8. Centrifugation time is established between the start of the speeding-up stage and the start of the braking stage. The start of the braking stage is audible. Stop the chronometer as soon as possible at this point.	--- --- --- --- 1228 rpm +/- 0.5% 540 seconds +/- 10 seconds

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DOCUMENT CHANGE HISTORY

Document change history			
Version	Date	Change	Rationale
2.0	2005/05/09	Document REGD-0000832 replaces document 57059V0M000	Uploaded document to GxPharma.
3.0	2011/04/28	Changed in step 5 the value of centrifuging time from 600 to 614 seconds. Changed in step 8 the value of centrifuging time from 10 minutes to 614 seconds.	When introducing 600 seconds in Centrifuge time field, the real value centrifuged is 614 seconds instead of 600 seconds. The value used to calculate the expected result should be 614 seconds.
3.0	2011/04/28	Updated the Document Change History: - Date format changed to YYYY-MM-DD - Dates changed to document version effective date.	Adoption of ISO 8601 date format to avoid ambiguity.
3.0	2011/04/28	ID-GELSTATION and PROVUE erased from the title.	Service manual intended for Grifols market.
4.0	July 2011	<i>Document REGD-0000832_v4 replaces document REGD-0000832_v3.</i>	Document version changed.
4.0	July 2011	Document title.	Commercial reference substituted by project reference.
4.0	July 2011	Document header.	Increase information to easy document search.
5.0	April 2018	Centrifugation parameters (speed and time).	Parameters did not match the ones checked during the OQ procedure.

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